

On Opial's Lemma

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Dedicated to the memory of Hedy Attouch, whose extraordinary mathematical vision, profound influence, and generous spirit continue to inspire our community

Abstract

Opial's Lemma is a fundamental result in the convergence analysis of sequences generated by optimization algorithms in real Hilbert spaces. We introduce the concept of Opial sequences—sequences for which the limit of the distance to each point in a given set exists. We systematically derive properties of Opial sequences, contrasting them with the well-studied Fejér monotone sequences, and establish conditions for weak and strong convergence. Key results include characterizations of weak convergence via weak cluster points (reaffirming Opial's Lemma), strong convergence via strong cluster points, and the behavior of projections onto Opial sets in terms of asymptotic centers. Special cases and examples are provided to highlight the subtle differences in convergence behaviour and projection properties compared to the Fejér monotone case.

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1 Introduction

Throughout this paper,

X is a real Hilbert space, with inner product $\langle \cdot, \cdot \rangle$ and induced norm $\| \cdot \|$. (1)

Many optimization problems boil down to finding a point in some solution set C , a subset of X . Often, a solution is found iteratively via some sequence $(x_n)_{n \in \mathbb{N}}$ generated by an optimization algorithm. The following result is key in the convergence analysis and part of the folklore:

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Fact 1.1 (Opial’s Lemma). *Let C be a nonempty subset of X and let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X such that*

$$(\forall c \in C) \quad \lim_n \|x_n - c\| \text{ exists.} \quad (2)$$

If all weak cluster points of $(x_n)_{n \in \mathbb{N}}$ lie in C , then $(x_n)_{n \in \mathbb{N}}$ converges weakly to some point in C .

We provide a proof of Fact 1.1 in Corollary 2.3 below. The ingredients for the proof of Fact 1.1 can be traced back to a classical paper by Opial [18]; however, the statement is not explicitly recorded there. For an explicit proof of Fact 1.1, see Peypouquet’s [19, Lemma 5.2]; however, he in turn gives credit to Baillon’s thesis (see [3, Chapitre 6]). Oftentimes, the sequence $(x_n)_{n \in \mathbb{N}}$ satisfies the even stronger property that $(x_n)_{n \in \mathbb{N}}$ is *Fejér monotone with respect to C* , i.e.,

$$(\forall c \in C)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - c\| \leq \|x_n - c\|. \quad (3)$$

In the Fejér monotonicity case, Fact 1.1 was proved by Browder in 1967 (see [11, Lemma 6]) and Browder states that it is essentially due Opial [18] and implicit in earlier work by Schäfer [23]. In many applications, C is typically the (nonempty) fixed point set of a nonexpansive mapping $T: X \rightarrow X$, i.e., $(\forall x \in X)(\forall y \in X) \|Tx - Ty\| \leq \|x - y\|$, and $(x_n)_{n \in \mathbb{N}} = (T^n x_0)_{n \in \mathbb{N}}$. In this case, $(T^n x_0)_{n \in \mathbb{N}}$ is indeed Fejér monotone with respect to $\text{Fix } T$. Many optimization algorithms fit this Fejér monotonicity framework; see, e.g., [6] which also contains basic properties of Fejér monotone sequences in [6, Section 5.1]. Nonetheless, the term Opial’s Lemma also refers precisely to Fact 1.1 — for recent instances, see, e.g., work by Attouch and Cabot [2, Lemma 2.5] and by Boţ, Csetnek, and Nguyen [10, Lemma A.3].

Somewhat surprisingly, sequences satisfying (2) have not been studied systematically. We thus introduce the following:

Definition 1.2 (Opial sequence and Opial set). *Let C be a nonempty subset of X . We say that a sequence $(x_n)_{n \in \mathbb{N}}$ in X is Opial with respect to C if*

$$(\forall c \in C) \quad \lim_n \|x_n - c\| \text{ exists.} \quad (4)$$

We occasionally refer to C as the corresponding Opial set.

Clearly, if $(x_n)_{n \in \mathbb{N}}$ is a sequence that is Fejér monotone with respect to C , then it is also Opial with respect to C .

We point out that various notions have been proposed that lie between Fejér monotone and Opial sequences — indeed, for a sequence $(x_n)_{n \in \mathbb{N}}$ in X and a nonempty subset C of X , we have:

- $(x_n)_{n \in \mathbb{N}}$ is *Fejér* monotone with respect to C* if

$$(\forall c \in C)(\exists N \in \mathbb{N})(\forall n \geq N) \quad \|x_{n+1} - c\| \leq \|x_n - c\|. \quad (5)$$

This notion was introduced by Behling, Bello-Cruz, Iusem, Liu, and Santos in [8] to analyze circumcenter methods, and further analyzed by Behling, Bello-Cruz, Iusem, Ribeiro, and Santos [9]. See also the preprint¹ [1] for an in-depth study of this notion.

¹ The preprint [1] was posted on arXiv about 9 months after the first version of this manuscript was posted and submitted. While this manuscript focuses on Opial sequences, the complementary preprint [1] deals with the relationships between Fejér, Fejér*, and quasi-Fejér sequences.

- $(x_n)_{n \in \mathbb{N}}$ is *quasi-Fejér monotone* with respect to C of
 - *type I* if $(\exists (\varepsilon_n)_{n \in \mathbb{N}} \in \ell_+^1)(\forall c \in C)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - c\| \leq \|x_n - c\| + \varepsilon_n$;
 - *type II* if $(\exists (\varepsilon_n)_{n \in \mathbb{N}} \in \ell_+^1)(\forall c \in C)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - c\|^2 \leq \|x_n - c\|^2 + \varepsilon_n$;
 - *type III* if $(\forall c \in C)(\exists (\varepsilon_n)_{n \in \mathbb{N}} \in \ell_+^1)(\forall n \in \mathbb{N}) \quad \|x_{n+1} - c\|^2 \leq \|x_n - c\|^2 + \varepsilon_n$.

The notions were introduced and systematically studied by Combettes in [14] to which we refer the reader for historical comments and applications. (See also the preprint [1].) Indeed, many optimization algorithms generate Fejér or at least quasi-Fejér sequences. Perhaps the most prominent example is the Krasnoselskii-Mann iteration, which generates a sequence $(x_n)_{n \in \mathbb{N}}$ via

$$x_{n+1} = (1 - \alpha_n)x_n + \alpha_n T x_n, \quad (7)$$

for a nonexpansive operator $T: X \rightarrow X$ and some suitably chosen sequence of relaxation parameters. One typically shows that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $\text{Fix } T$ and that all weak cluster points of $(x_n)_{n \in \mathbb{N}}$ lie in $\text{Fix } T$, which yields weak convergence of $(x_n)_{n \in \mathbb{N}}$ thanks to Fact 1.1.

It is clear that Fejér* monotone sequences are Opial. The fact that every quasi-Fejér monotone sequence is Opial is known (see [14, Section 3]) and a consequence of the following:

Fact 1.3 (Robbins-Siegmund).² *Let $(\alpha_n)_{n \in \mathbb{N}}, (\beta_n)_{n \in \mathbb{N}}, (\delta_n)_{n \in \mathbb{N}}, (\varepsilon_n)_{n \in \mathbb{N}}$ be sequences of nonnegative real numbers such that $\sum_{n \in \mathbb{N}} \delta_n < +\infty$ and $\sum_{n \in \mathbb{N}} \varepsilon_n < +\infty$ and*

$$(\forall n \in \mathbb{N}) \quad \alpha_{n+1} \leq (1 + \delta_n)\alpha_n - \beta_n + \varepsilon_n \quad (8)$$

Then $(\alpha_n)_{n \in \mathbb{N}}$ is convergent and $\sum_{n \in \mathbb{N}} \beta_n < +\infty$.

Let us point out that there exist sequences that are Opial but are neither Fejér* monotone nor quasi-Fejér monotone:

Example 1.4. *Suppose that $X = \mathbb{R}$ and set*

$$(x_n)_{n \in \mathbb{N}} := \left(0, \frac{1}{\sqrt{1}}, 0, \frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{3}}, \dots\right). \quad (9)$$

Then $x_n \rightarrow 0$, $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $\{0\}$, but $(x_n)_{n \in \mathbb{N}}$ is neither Fejér monotone nor quasi-Fejér monotone with respect to $\{0\}$ (because $\sum_{n \geq 1} 1/\sqrt{n}$ diverges to $+\infty$).*

Having seen various notions between Fejér monotone and Opial sequences, we now state that the goal of this paper is to derive useful properties of Opial sequences and contrast them to properties of Fejér monotone sequences.

We now summarize some key results in the following table. Here $(x_n)_{n \in \mathbb{N}}$ is a sequence in X and C is a nonempty subset of X . We denote the sets of weak and strong cluster points of $(x_n)_{n \in \mathbb{N}}$ by $\mathcal{W}$ and $\mathcal{S}$, respectively. When C is closed and convex, then we denote the strong cluster points of $(P_C x_n)_{n \in \mathbb{N}}$ by $\mathcal{S}((P_C x_n)_{n \in \mathbb{N}})$; in addition, the (well-defined) asymptotic center (see Definition 3.1 below) of $(x_n)_{n \in \mathbb{N}}$ with respect to C is denoted by $\hat{c}$. Let us explain how one reads

²Fact 1.3 is a non-stochastic version of a classical probabilistic result by Robbins and Siegmund [22]. See also [20, Lemma 11 on page 50], [14, Lemma 3.1], [6, Lemma 5.31], as well as [17, Theorem 3.2] for a very recent quantitative version.

the table: The first column lists a possible additional assumption. The second and third column present the strongest conclusions under this extra assumption. We write “same” in the third column if the Opial conclusion is identical to the Fejér conclusion, and “when” when the additional assumption is required to reach the same conclusion. The fourth column proves pointers to the relevant conclusion in the Opial case.

Assumption	$(x_n)_{n \in \mathbb{N}}$ is Fejér wrt C	$(x_n)_{n \in \mathbb{N}}$ is Opial wrt C	Notes
none	$(\mathcal{W} - \mathcal{W}) \subseteq (C - C)^\perp$	same	Theorem 2.2
$\mathcal{W} \subseteq C$	$x_n \rightarrow$ some point in C	same	Corollary 2.3
$\mathcal{S} \cap C \neq \emptyset$	$x_n \rightarrow$ some point in C	same	Proposition 2.4
$\text{int } C \neq \emptyset$	$x_n \rightarrow$ some point in X	$x_n \rightarrow$ some point in X	Corollary 2.6
none	$(x_n)_{n \in \mathbb{N}}$ is Fejér wrt $\overline{\text{conv}} C$	$(x_n)_{n \in \mathbb{N}}$ is Opial wrt $\text{aff } C$	Proposition 2.8
$C = X$	$(x_n)_{n \in \mathbb{N}} = (x_0)_{n \in \mathbb{N}}$	$x_n \rightarrow$ some point in X	Corollary 2.11
$C = \overline{\text{conv}} C$	$P_C x_n \rightarrow \hat{c}$	$P_C x_n \rightarrow \hat{c}$ when $d_C(x_n) \rightarrow 0$	Theorem 3.5
$C = \overline{\text{conv}} C$	$P_C x_n \rightarrow \hat{c}$	$\mathcal{S}((P_C x_n)_{n \in \mathbb{N}}) \subseteq \{\hat{c}\}$	Proposition 3.7
$C = \overline{\text{conv}} C$	$P_C x_n \rightarrow \hat{c}$	when C is affine	Theorem 3.12
$C = \overline{\text{conv}} C$	$(d_C(x_n))_{n \in \mathbb{N}}$ is convergent	when $\overline{\{x_n\}_{n \in \mathbb{N}}}$ is compact	Proposition 3.10

The sharpness of our results is illustrated by various examples provided in this paper.

The remainder of the paper is organized as follows. In [Section 2](#), we provide basic properties of Opial sequences. After reviewing results on the asymptotic center, we study the projections of Opial sequences in [Section 3](#). Finally, the notation we employ is fairly standard and largely follows [\[6\]](#) to which we also refer the reader for background material.

2 Weak and strong convergence of Opial sequences

We now systematically derive properties of Opial sequences (which we defined in [Definition 1.2](#)).

Proposition 2.1 (basic properties). *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty subset C of X . Then the following hold:*

- (i) $(x_n)_{n \in \mathbb{N}}$ is bounded.
- (ii) $\overline{\lim}_n d_C(x_n) \leq \inf_{c \in C} \lim_n \|x_n - c\|$.

Proof. (i): By definition, for every $c \in C$, the $\lim_n \|x_n - c\|$ exists. Hence $(\|x_n - c\|)_{n \in \mathbb{N}}$ is bounded and thus $(x_n)_{n \in \mathbb{N}}$ is bounded.

(ii): Let $c \in C$. Then $(\forall n \in \mathbb{N}) d_C(x_n) \leq \|x_n - c\|$. Hence $\overline{\lim}_n d_C(x_n) \leq \overline{\lim}_n \|x_n - c\| = \lim_n \|x_n - c\|$. Now take the infimum over $c \in C$. ■

Throughout this paper, we will repeatedly use the fact that every Opial sequence is bounded. Because of this, we are interested to learn about the location of weak cluster points:

Theorem 2.2 (location of weak cluster points). *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty subset C of X . Let w_1 and w_2 be two weak cluster points of $(x_n)_{n \in \mathbb{N}}$. Then*

$$w_1 - w_2 \in (C - C)^\perp. \quad (10)$$

Proof. (We follow the proof for Fejér monotone sequences presented in [4, Theorem 6.2.2(ii)].) By **Proposition 2.1(i)**, $(x_n)_{n \in \mathbb{N}}$ is bounded. Hence there exists subsequences $(x_{k_n})_{n \in \mathbb{N}}$ and $(x_{l_n})_{n \in \mathbb{N}}$ of $(x_n)_{n \in \mathbb{N}}$ such that

$$x_{k_n} \rightharpoonup w_1 \text{ and } x_{l_n} \rightharpoonup w_2. \quad (11)$$

Let c_1 and c_2 be in C . Because $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C , we have

$$\lim_n \|x_n - c_1\| \text{ and } \lim_n \|x_n - c_2\| \text{ exist.}$$

Hence $(\|x_n - c_1\|^2)_{n \in \mathbb{N}}$ and $(\|x_n - c_2\|^2)_{n \in \mathbb{N}}$ are convergent sequences, and so is their difference

$$(\|x_n - c_1\|^2 - \|x_n - c_2\|^2)_{n \in \mathbb{N}}$$

After expanding and simplifying, we deduce that $(-2 \langle x_n, c_1 - c_2 \rangle)_{n \in \mathbb{N}}$ is convergent, and so

$$\lim_n \langle x_n, c_1 - c_2 \rangle \text{ exists.} \quad (12)$$

Thus, taking the limit in (12) along the subsequences from (11), we obtain

$$\langle w_1, c_1 - c_2 \rangle = \langle w_2, c_1 - c_2 \rangle.$$

Therefore, $\langle w_1 - w_2, c_1 - c_2 \rangle = 0$, as claimed. ■

Corollary 2.3 (Opial's Lemma). *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty subset C of X . Then: $(x_n)_{n \in \mathbb{N}}$ converges weakly to some point in $C \Leftrightarrow$ all weak cluster points of $(x_n)_{n \in \mathbb{N}}$ lie in C .*

Proof. “ $\Rightarrow$ ”: Clear. “ $\Leftarrow$ ”: By **Proposition 2.1(i)**, the sequence $(x_n)_{n \in \mathbb{N}}$ is bounded. Let w_1 and w_2 be two weak cluster points of $(x_n)_{n \in \mathbb{N}}$. By assumption, $\{w_1, w_2\} \subseteq C$ and so $w_1 - w_2 \in C - C$. On the other hand, by **Theorem 2.2**, $w_1 - w_2 \in (C - C)^\perp$. Altogether $w_1 - w_2 \in (C - C) \cap (C - C)^\perp = \{0\}$ and so $w_1 = w_2$. ■

Corollary 2.3 provides a convenient characterization of weak convergence to an element in the Opial set. We now present the counterpart for strong convergence:

Proposition 2.4. *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty subset C of X . Then: $(x_n)_{n \in \mathbb{N}}$ converges strongly to some point in $C \Leftrightarrow$ there exists a cluster point of $(x_n)_{n \in \mathbb{N}}$ in C .*

Proof. “ $\Rightarrow$ ”: Clear. “ $\Leftarrow$ ”: Suppose c is a cluster point of $(x_n)_{n \in \mathbb{N}}$ that lies in C . Then there exists a subsequence $(x_{k_n})_{n \in \mathbb{N}}$ of $(x_n)_{n \in \mathbb{N}}$ such that

$$x_{k_n} \rightarrow c \in C.$$

Because $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C , the $\ell := \lim_{n \rightarrow \infty} \|x_n - c\|$ exists. Of course, $\|x_{k_n} - c\| \rightarrow \ell$ as well. But $x_{k_n} - c \rightarrow 0$, so $\ell = 0$. It follows that $\|x_n - c\| \rightarrow 0$ and therefore $x_n \rightarrow c$. ■

Corollary 2.5. *Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to X and such that $\{x_n\}_{n \in \mathbb{N}}$ is relatively compact, i.e., $\{x_n\}_{n \in \mathbb{N}}$ is compact. Then $(x_n)_{n \in \mathbb{N}}$ is strongly convergent.*

Proof. The relative compactness assumption guarantees that $(x_n)_{n \in \mathbb{N}}$ has a strong cluster point. Now apply **Proposition 2.4** with $C = X$. ■

Loosely speaking, the next result says that if the Opial set C is “sufficiently large”, then the Opial sequence converges weakly:

Corollary 2.6. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty subset C of X . If $\overline{\text{aff } C} = X$, then $(x_n)_{n \in \mathbb{N}}$ converges weakly to a point in X .

Proof. By assumption, the closed linear span of $C - C$ is equal to entire space X . But then $(C - C)^\perp = X^\perp = \{0\}$ and so all weak cluster points of $(x_n)_{n \in \mathbb{N}}$ must coincide by [Theorem 2.2](#). ■

Remark 2.7. Assume that a sequence $(x_n)_{n \in \mathbb{N}}$ in X is Opial with respect to a nonempty subset C of X and that one of the following holds:

- (i) $\text{int } C \neq \emptyset$, or
- (ii) $X = \ell^2$ and $C = \ell^2_+$ (for which $\text{int } C = \emptyset$).

Then $(x_n)_{n \in \mathbb{N}}$ is weakly convergent because each condition guarantees that $\overline{\text{aff } C} = X$ and [Corollary 2.6](#) applies. We also note that [Corollary 2.6](#) is known in the context of Fejér monotonicity (see [\[13, Proposition 3.7\(iii\)\]](#)).

The next result is somewhat surprising: the Opial set of an Opial sequence can be enlarged to a closed affine subspace!

Proposition 2.8. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty subset C of X . Then $(x_n)_{n \in \mathbb{N}}$ is also Opial with respect to $\overline{\text{aff } C}$.

Proof. For $c \in C$, set $\ell(c) := \lim_n \|x_n - c\|$ which is well defined by assumption.

Step 1: Extension to the affine hull.

We first show that the Opial set C can be extended to $\text{aff } C$. Pick a finite index set I and take $c_i \in C$ for every $i \in I$. Also, let each λ_i be real, with $\sum_{i \in I} \lambda_i = 1$. Using [\[6, Lemma 2.14\(ii\)\]](#), we see that when $n \rightarrow \infty$, we obtain

$$\begin{aligned} \left\| x_n - \sum_{i \in I} \lambda_i c_i \right\|^2 &= \left\| \sum_{i \in I} \lambda_i (x_n - c_i) \right\|^2 \\ &= \sum_{i \in I} \lambda_i \|x_n - c_i\|^2 - \frac{1}{2} \sum_{i \in I, j \in I} \lambda_i \lambda_j \|(x_n - c_i) - (x_n - c_j)\|^2 \\ &= \sum_{i \in I} \lambda_i \|x_n - c_i\|^2 - \frac{1}{2} \sum_{i \in I, j \in I} \lambda_i \lambda_j \|c_i - c_j\|^2 \\ &\rightarrow \sum_{i \in I} \lambda_i \ell^2(c_i) - \frac{1}{2} \sum_{i \in I, j \in I} \lambda_i \lambda_j \|c_i - c_j\|^2. \end{aligned}$$

as claimed³.

Step 2: Extension to the closure.

Our proof follows along lines of the proof of [\[9, Proposition 3.2\(i\)\]](#). Let $\bar{c} \in \overline{C}$ say $c_m \rightarrow \bar{c}$, where each c_m lies in C . The triangle inequality yields the key inequality

$$(\forall m \in \mathbb{N})(\forall n \in \mathbb{N}) \quad -\|c_m - \bar{c}\| \leq \|x_n - \bar{c}\| - \|x_n - c_m\| \leq \|c_m - \bar{c}\|,$$

from which the result will readily follow: indeed, keeping m fixed and taking $\liminf/\limsup$ as $n \rightarrow \infty$, we deduce that

$$-\|c_m - \bar{c}\| \leq \varliminf_n \|x_n - \bar{c}\| - \ell(c_m) \leq \overline{\lim_n \|x_n - \bar{c}\|} - \ell(c_m) \leq \|c_m - \bar{c}\|.$$

³In passing, we point out that $\ell^2(\sum_i \lambda_i c_i) = \sum_{i \in I} \lambda_i \ell^2(c_i) - \frac{1}{2} \sum_{i \in I, j \in I} \lambda_i \lambda_j \|c_i - c_j\|^2$.

Letting now $m \rightarrow \infty$ shows that $\lim_n \|x_n - \bar{c}\| = \overline{\lim_n \|x_n - \bar{c}\|} = \lim_m \ell(c_m)$. Hence $(\|x_n - \bar{c}\|)_{n \in \mathbb{N}}$ converges and $\ell(\bar{c}) = \lim_n \|x_n - \bar{c}\| = \lim_m \ell(c_m)$ exists⁴.

Now combine Step 1 and Step 2 to obtain the conclusion. ■

Remark 2.9. Some comments on [Proposition 2.8](#) are in order.

(i) For a nonempty subset C of X , we have the equivalences:

$$(x_n)_{n \in \mathbb{N}} \text{ is Opial with respect to } C \Leftrightarrow (x_n)_{n \in \mathbb{N}} \text{ is Opial with respect to } \overline{C} \quad (13a)$$

$$\Leftrightarrow (x_n)_{n \in \mathbb{N}} \text{ is Opial with respect to } \overline{\text{aff } C}. \quad (13b)$$

(ii) The counterpart for Fejér monotone sequence states that if $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C , then $(x_n)_{n \in \mathbb{N}}$ is also Fejér monotone with respect to $\overline{\text{conv}} C$. This is part of the folklore; see, e.g., [7, Lemma 2.1(ii)].

Example 2.10. Suppose that $X = \mathbb{R}^2$, and define $(x_n)_{n \in \mathbb{N}} := ((-1)^n, 0)_{n \in \mathbb{N}}$. Clearly, $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $C := \{0\} \times [-1, 1]$. By [Proposition 2.8](#), $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $\text{aff } C = \{0\} \times \mathbb{R}$. Note that there is no proper superset D of $\{0\} \times \mathbb{R}$ such that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to D . This illustrates the sharpness of [Proposition 2.8](#).

In the following two results, we describe precisely the relationship between sequences that are Opial with respect to X , and weakly and strongly convergent sequences.

Corollary 2.11. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to X . Then $(x_n)_{n \in \mathbb{N}}$ is weakly convergent.

Proof. Clear from [Corollary 2.3](#) with $C = X$. ■

Proposition 2.12. Let $(x_n)_{n \in \mathbb{N}}$ be a weakly convergent sequence in X . Then $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $X \Leftrightarrow (\|x_n\|)_{n \in \mathbb{N}}$ is convergent.

Proof. Suppose $x_n \rightharpoonup \bar{x}$. Let's fix $x \in X$, write

$$\|x_n - x\|^2 - \|x_n\|^2 = \|x\|^2 - 2 \langle x_n, x \rangle \rightarrow \|x\|^2 - 2 \langle \bar{x}, x \rangle,$$

which shows that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $X \Leftrightarrow \lim_n \|x_n\|$ exists ($\Leftrightarrow (x_n)_{n \in \mathbb{N}}$ is Opial with respect to $\{0\}$). ■

Theorem 2.13. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . Then $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $X \Leftrightarrow (x_n)_{n \in \mathbb{N}}$ is weakly convergent and $(\|x_n\|)_{n \in \mathbb{N}}$ is convergent.

Proof. “ $\Rightarrow$ ”: [Corollary 2.11](#) and [Proposition 2.12](#). “ $\Leftarrow$ ”: [Proposition 2.12](#). ■

Corollary 2.14. Suppose that X is finite-dimensional and let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . Then $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to $X \Leftrightarrow (x_n)_{n \in \mathbb{N}}$ is convergent.

Proof. Recall that because X is finite-dimensional, strong and weak convergence for sequences coincide. “ $\Rightarrow$ ”: Clear from [Theorem 2.13](#). “ $\Leftarrow$ ”: This is clear because every strongly convergent sequence is Opial with respect to X . ■

⁴We note the “continuity property” $c_m \rightarrow \bar{c} \Rightarrow \ell(c_m) \rightarrow \ell(\bar{c})$.

Example 2.15. Suppose that $X = \ell^2$ and let $(e_n)_{n \in \mathbb{N}}$ be the sequence of standard unit vectors in X . Then:

- (i) $(e_n)_{n \in \mathbb{N}}$ is Opial with respect to X , $e_n \rightarrow 0$; however, $e_n \not\rightarrow 0$ and $(e_n)_{n \in \mathbb{N}}$ is Fejér only with respect to $\{c = (\gamma_n) \in X \mid (\gamma_n)_{n \in \mathbb{N}} \text{ is increasing}\}$.
- (ii) $(x_n)_{n \in \mathbb{N}} = (e_0, 0, e_1, 0, e_2, 0, \dots)$ converges weakly to 0; however $(x_n)_{n \in \mathbb{N}}$ is not Opial (and hence not Fejér monotone) with respect to any nonempty subset of X .

Proof. (i): Clearly, $e_n \rightarrow 0$ and $\|e_n\| \equiv 1$. By Theorem 2.13, $(e_n)_{n \in \mathbb{N}}$ is Opial with respect to X . Suppose that $(e_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to $\{c\}$, where $c = (\gamma_n)_{n \in \mathbb{N}} \in X$. Then $(\forall n \in \mathbb{N}) \ \|e_{n+1} - c\| \leq \|e_n - c\| \Leftrightarrow \|e_{n+1} - c\|^2 \leq \|e_n - c\|^2 \Leftrightarrow 1 - 2\langle e_{n+1}, c \rangle + \|c\|^2 \leq 1 - 2\langle e_n, c \rangle + \|c\|^2 \Leftrightarrow \langle e_n - e_{n+1}, c \rangle \leq 0 \Leftrightarrow \gamma_n - \gamma_{n+1} \leq 0 \Leftrightarrow \gamma_n \leq \gamma_{n+1}$, as announced.

(ii): Clearly, $x_n \rightarrow 0$. However, $(\|x_n\|)_{n \in \mathbb{N}} = (1, 0, 1, 0, 1, 0, \dots)$ is not convergent. By Theorem 2.13, $(x_n)_{n \in \mathbb{N}}$ is not Opial with respect to X . Suppose to the contrary that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C and let $c \in C$. Then $\ell := \lim_n \|x_n - c\|$ exists. Considering $(x_{2n+1})_{n \in \mathbb{N}}$, we see that $\ell = \|c\|$. On the other hand, considering $(x_{2n})_{n \in \mathbb{N}}$, we have $\|c\|^2 \leftarrow \|x_{2n} - c\|^2 = 1 + \|c\|^2 - 2\langle x_{2n}, c \rangle \rightarrow 1 + \|c\|^2$, which is absurd. ■

Remark 2.16. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X . Then $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to $X \Leftrightarrow (x_n)_{n \in \mathbb{N}} = (x_0)_{n \in \mathbb{N}}$ is a constant sequence. Example 2.15(i) illustrates a striking difference between Opial and Fejér monotone sequences: if $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to a subset C of X with $\text{int } C \neq \emptyset$, then a result due to Raik (see [21] or [6, Proposition 5.10]) implies that $(x_n)_{n \in \mathbb{N}}$ converges strongly with the “finite-length” property $\sum_n \|x_n - x_{n+1}\| < +\infty$. In contrast, the sequence $(e_n)_{n \in \mathbb{N}}$ from Example 2.15(i), which is Opial with respect to X , converges only weakly!

3 Opial sequences and projections onto Opial sets

In this section, we study the behaviour of $(P_C x_n)$ when $(x_n)_{n \in \mathbb{N}}$ is a sequence in X such that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to the nonempty closed convex subset C of X . The closedness and convexity of C is imposed because our analysis relies on the following concept, introduced by Edelstein [15] in 1972 originally for analyzing nonexpansive mappings (see also [16, Chapter 9]):

Definition 3.1 (asymptotic center). Let $(u_n)_{n \in \mathbb{N}}$ be a bounded sequence in X and let C be a nonempty closed convex subset of X . Then there is a unique point $\hat{c} \in C$ such that

$$(\forall c \in C \setminus \{\hat{c}\}) \quad \overline{\lim}_n \|u_n - \hat{c}\| < \overline{\lim}_n \|u_n - c\|. \quad (14)$$

The point $\hat{c}$ is the asymptotic center of $(u_n)_{n \in \mathbb{N}}$ with respect to C , and we write $\hat{c} = A_C(u_n)_{n \in \mathbb{N}}$ or, more succinctly, $A_C(u_{\mathbb{N}})$.

We start with the following simple observation.

Lemma 3.2. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty closed convex subset C of X . Let $(x_{k_n})_{n \in \mathbb{N}}$ be an arbitrary subsequence of $(x_n)_{n \in \mathbb{N}}$. Then these two sequences have the same asymptotic center with respect to C , i.e., $A_C(x_{\mathbb{N}}) = A_C(x_{k_{\mathbb{N}}})$.

Proof. Recall that $(x_n)_{n \in \mathbb{N}}$ is bounded by Proposition 2.1(i). Let $c \in C$. Because $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C , $(\|x_n - c\|)_{n \in \mathbb{N}}$ is convergent and $\lim_n \|x_n - c\| = \lim_n \|x_{k_n} - c\|$. Thus $\overline{\lim}_n \|x_n - c\| = \overline{\lim}_n \|x_{k_n} - c\|$ and so the asymptotic centers coincide. ■

We also have the following properties:

Fact 3.3 (asymptotic center). (See [6, Proposition 11.18] and [12, Theorem 3.3.2].) Let $(u_n)_{n \in \mathbb{N}}$ be a bounded sequence in X and let C be a nonempty closed convex subset of X . Set $f: X \rightarrow \mathbb{R}: x \mapsto \overline{\lim}_n \|x - u_n\|^2$. Then the following hold:

- (i) The function $f + \iota_C$ is supercoercive, and strongly convex with constant 2.
- (ii) $A_C(u_{\mathbb{N}})$, the asymptotic center of $(u_n)_{n \in \mathbb{N}}$ with respect to C , is the unique minimizer of $f + \iota_C$.
- (iii) If $u_n \rightarrow u$, then $(\forall x \in X) f(x) = \|x - u\|^2 + f(u)$ and $P_C u = A_C(u_{\mathbb{N}})$.
- (iv) If $(u_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C , then $P_C u_n \rightarrow A_C(u_{\mathbb{N}})$.

Corollary 3.4. Let C be a nonempty closed convex subset of X , and let $(c_n)_{n \in \mathbb{N}}$ be a sequence in C that is Opial with respect to C . Then $c_n \rightarrow A_C(c_{\mathbb{N}})$.

Proof. By Proposition 2.1(i), the sequence $(c_n)_{n \in \mathbb{N}}$ is bounded. Let $\bar{c} \in C$ be a weak cluster point of $(c_n)_{n \in \mathbb{N}}$, say $c_{k_n} \rightharpoonup \bar{c}$. On the one hand, Fact 3.3(iii) yields $\bar{c} = P_C \bar{c} = A_C(c_{k_{\mathbb{N}}})$. On the other hand, $A_C(c_{\mathbb{N}}) = A_C(c_{k_{\mathbb{N}}})$ by Lemma 3.2. Altogether, $\bar{c} = A_C(c_{\mathbb{N}})$ which yields the result. ■

Theorem 3.5. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty closed convex subset C of X and such that $d_C(x_n) \rightarrow 0$. Then the asymptotic centers of $(x_n)_{n \in \mathbb{N}}$ and $(P_C x_n)_{n \in \mathbb{N}}$ with respect to C coincide, and $P_C x_n \rightarrow A_C(x_{\mathbb{N}})$.

Proof. By Proposition 2.1(i), $(x_n)_{n \in \mathbb{N}}$ is bounded. Because P_C is nonexpansive, it follows that $(P_C x_n)_{n \in \mathbb{N}}$ is bounded as well. Because $x_n - P_C x_n \rightarrow 0$, we see that the inner product term in

$$(\forall c \in C)(\forall n \in \mathbb{N}) \quad \|x_n - c\|^2 = \|x_n - P_C x_n\|^2 + \|P_C x_n - c\|^2 + 2 \langle x_n - P_C x_n, P_C x_n - c \rangle$$

is controlled and goes to 0. Now $(\|x_n - c\|^2)_{n \in \mathbb{N}}$ converges, and therefore $(\|P_C x_n - c\|^2)_{n \in \mathbb{N}}$ converges as well; moreover,

$$(\forall c \in C) \quad \lim_n \|x_n - c\| = \lim_n \|P_C x_n - c\|. \quad (15)$$

This implies that $A_C(x_{\mathbb{N}}) = A_C(P_C x_{\mathbb{N}})$ and that $(P_C x_n)_{n \in \mathbb{N}}$ is Opial with respect to C . By Corollary 3.4, $P_C x_n \rightarrow A_C(P_C x_{\mathbb{N}})$ and we are done. ■

Remark 3.6. Let us contrast Theorem 3.5 with its Fejér counterpart: if $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to a nonempty closed convex subset C of X , then Fact 3.3(iv) yields the strong convergent statement $P_C x_n \rightarrow A_C(x_{\mathbb{N}})$ even without the assumption that $d_C(x_n) \rightarrow 0$. On the other hand, Example 2.15(i) illustrates that the convergence in Theorem 3.5 may fail to be strong: $P_X e_n = e_n \rightarrow 0$ but $e_n \not\rightarrow 0$. Hence Theorem 3.5 is sharp in the sense that it is impossible to obtain strong convergence of $(P_C x_n)_{n \in \mathbb{N}}$ even when $d_C(x_n) \rightarrow 0$.

Proposition 3.7. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty closed convex subset C of X . Then all strong cluster points of $(P_C x_n)_{n \in \mathbb{N}}$ coincide with $A_C(x_{\mathbb{N}})$.

Proof. Suppose that $P_C x_{k_n} \rightarrow \bar{c}$. Then $(\forall c \in C) \|x_{k_n} - \bar{c}\| \leq \|x_{k_n} - P_C x_{k_n}\| + \|P_C x_{k_n} - \bar{c}\| \leq \|x_{k_n} - c\| + \|P_C x_{k_n} - \bar{c}\|$. Because $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C , we learn with the help of Lemma 3.2 that

$$(\forall c \in C) \quad \lim_n \|x_n - \bar{c}\| = \lim_n \|x_{k_n} - \bar{c}\| \leq \lim_n \|x_{k_n} - c\| = \lim_n \|x_n - c\|;$$

therefore, $\bar{c} = A_C(x_{\mathbb{N}})$. ■

Corollary 3.8. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty closed convex subset C of X . Suppose that $\{P_C x_n\}_{n \in \mathbb{N}}$ is relatively compact. Then $P_C x_n \rightarrow A_C(x_{\mathbb{N}})$.

Proof. Clear from [Proposition 3.7](#). ■

Corollary 3.9. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty closed convex subset C of X . Suppose that $\{x_n\}_{n \in \mathbb{N}}$ is relatively compact, that C is boundedly compact⁵, or that X is finite-dimensional. Then $P_C x_n \rightarrow A_C(x_{\mathbb{N}})$.

Proof. Each assumption implies that $\{P_C x_n\}_{n \in \mathbb{N}}$ is relatively compact, and the conclusion thus follows from [Corollary 3.8](#). ■

Provided a relative compactness condition holds, we now have a result that sharpens [Proposition 2.1\(ii\)](#):

Proposition 3.10. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a nonempty closed convex subset C of X . Suppose that $\{P_C x_n\}_{n \in \mathbb{N}}$ is relatively compact. Then $(d_C(x_n))_{n \in \mathbb{N}}$ is convergent and $\lim_n d_C(x_n) = \lim_n \|x_n - A_C(x_{\mathbb{N}})\| = \min_{c \in C} \lim_n \|x_n - c\|$.

Proof. Indeed, for all $n \in \mathbb{N}$, we have:

$$\begin{aligned} d_C^2(x_n) &= \|x_n - P_C(x_n)\|^2 \\ &= \|x_n - A_C(x_{\mathbb{N}})\|^2 + \|A_C(x_{\mathbb{N}}) - P_C(x_n)\|^2 + 2\langle x_n - A_C(x_n), A_C(x_{\mathbb{N}}) - P_C(x_n) \rangle. \end{aligned}$$

However, [Corollary 3.8](#) implies that the last two summands tend to zero as $\{P_C x_n\}_{n \in \mathbb{N}}$ is assumed to be relatively compact. The conclusion thus follows because $\lim_n \|x_n - A_C(x_{\mathbb{N}})\|$ in view of the Opial assumption with respect to C . ■

Remark 3.11. Consider [Proposition 3.10](#). Without the relative compactness assumption, the conclusion that $\lim_n d_C(x_n) = \lim_n \|x_n - A_C(x_{\mathbb{N}})\|$ may fail: indeed, let's revisit [Example 2.15\(i\)](#), where $(e_n)_{n \in \mathbb{N}}$ is Opial with respect to X ; hence, $d_X(e_n) \equiv 0$ while $A_X(e_{\mathbb{N}}) = 0$ (by [Fact 3.3\(iii\)](#)) and so $\|e_n - A_C(e_{\mathbb{N}})\| \equiv 1$. In contrast, if $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to C , then $P_C x_n \rightarrow A_C(x_{\mathbb{N}})$ by [Fact 3.3\(iv\)](#) and thus $\lim_n d_C(x_n) = \lim_n \|x_n - P_C x_n\| = \lim_n \|x_n - A_C(x_{\mathbb{N}})\|$ without any relative compactness assumption.

We now show that if the Opial set is affine, then the projected sequence is weakly convergent:

Theorem 3.12 (projections onto an affine Opial set). Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X that is Opial with respect to a closed affine subspace C of X . Then $(P_C x_n)_{n \in \mathbb{N}}$ is weakly convergent to $A_C(x_{\mathbb{N}})$.

Proof. First, let $Y := C - C$ be the closed linear subspace parallel to C . Then $P_C: x \mapsto z + P_Y x$, where $z := P_C 0 \in Y^\perp$. Next, let c_1, c_2 be two weak cluster points of $(P_C x_n)_{n \in \mathbb{N}}$, say

$$P_C x_{k_n} \rightharpoonup c_1 \text{ and } P_C x_{l_n} \rightharpoonup c_2.$$

After passing to subsequences and re-labelling if necessary, we additionally assume that $x_{k_n} \rightharpoonup w_1$ and $x_{l_n} \rightharpoonup w_2$. Because P_Y is weakly continuous ([\[6, Proposition 4.19\(i\)\]](#)), we deduce that P_C is

⁵Recall that C is *boundedly compact* if the intersection of C with every closed ball is compact.

weakly continuous as well and thus $P_C x_{k_n} \rightharpoonup c_1 = P_C w_1 = z + P_Y w_1$ and $P_C x_{l_n} \rightharpoonup c_2 = P_C w_2 = z + P_Y w_2$. Hence

$$c_1 - c_2 = P_C w_1 - P_C w_2 = P_Y(w_1 - w_2). \quad (16)$$

Furthermore, by [Theorem 2.2](#),

$$w_1 - w_2 \in (C - C)^\perp = Y^\perp. \quad (17)$$

Combining (16) with (17), we deduce that $c_1 - c_2 = P_Y(w_1 - w_2) \in P_Y(Y^\perp) = \{0\}$ and so $c_1 = c_2$. We've shown that $(P_C x_n)_{n \in \mathbb{N}}$ is weakly convergent to $\bar{c} := c_1 = c_2$.

Next, on the one hand, by [Lemma 3.2](#), $A_C(x_{\mathbb{N}}) = A_C(x_{k_{\mathbb{N}}})$. On the other hand, by [Fact 3.3\(iii\)](#), $\bar{c} = c_1 = P_C w_1 = A_C(x_{k_{\mathbb{N}}})$. Altogether, $\bar{c} = A_C(x_{\mathbb{N}})$. $\blacksquare$

Example 3.13. Suppose that $X = \ell^2(\mathbb{N})$, with the standard basis of unit vectors $(e_n)_{n \in \mathbb{N}}$. Define the sequence $(x_n)_{n \in \mathbb{N}}$ in X by

$$(\forall n \in \mathbb{N}) \quad x_n := \begin{cases} e_0 + e_1, & \text{if } n \text{ is even;} \\ e_1 + e_{n+1}, & \text{if } n \text{ is odd.} \end{cases} \quad (18)$$

Then the following hold:

- (i) Set $Y := \{e_0\}^\perp$. The sequence $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to Y , but not with respect to X . The sequence $(P_Y x_n)_{n \in \mathbb{N}} = (e_1, e_1 + e_2, e_1, e_1 + e_4, e_1, e_1 + e_6, \dots)$ converges weakly to $e_1 = A_Y(x_{\mathbb{N}})$ but $(P_Y x_n)_{n \in \mathbb{N}}$ is not Opial with respect to Y . The sequence $(x_n - P_Y x_n)_{n \in \mathbb{N}} = (e_0, 0, e_0, 0, \dots)$ does not converge weakly and the sequence $(d_Y(x_n))_{n \in \mathbb{N}} = (1, 0, 1, 0, \dots)$ is not convergent either.
- (ii) Set $B := \{x \in X \mid \|x\| \leq 1\}$ and $C := B \cap Y$. The sequence $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C . But $(P_C x_n)_{n \in \mathbb{N}} = \frac{1}{\sqrt{2}}(\sqrt{2}e_1, e_1 + e_2, \sqrt{2}e_1, e_1 + e_4, \sqrt{2}e_1, e_1 + e_6, \dots)$ is not weakly convergent and $d_C(x_n) = (1, \sqrt{2} - 1, 1, \sqrt{2} - 1, \dots)$ is not convergent either.

Proof. Observe first that $(x_n)_{n \in \mathbb{N}}$ has exactly one strong cluster point, namely $e_0 + e_1$ and exactly one different weak cluster point, namely e_1 . In particular, $(x_n)_{n \in \mathbb{N}}$ is not weakly convergent.

(i): Let $n \in \mathbb{N}$ and $y \in Y$, i.e., $y \perp e_0$. If n is even, we have $\|x_n - y\|^2 = \|(e_0 + e_1) - y\|^2 = \|e_0 + (e_1 - y)\|^2 = \|e_0\|^2 + \|e_1 - y\|^2 = 1 + \|e_1 - y\|^2$. And if n is odd, we have $\|x_n - y\|^2 = \|(e_1 + e_{n+1}) - y\|^2 = \|(e_1 - y) + e_{n+1}\|^2 = \|e_1 - y\|^2 + \|e_{n+1}\|^2 + 2\langle e_1 - y, e_{n+1} \rangle = \|e_1 - y\|^2 + 1 + 2\langle e_1 - y, e_{n+1} \rangle \rightarrow \|e_1 - y\|^2 + 1$ because $e_{n+1} \rightharpoonup 0$. It follows that

$$\|x_n - y\|^2 \rightarrow \|e_1 - y\|^2 + 1, \quad (19)$$

and so $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to Y . If $(x_n)_{n \in \mathbb{N}}$ were Opial with respect to X , then $(x_n)_{n \in \mathbb{N}}$ would be weakly convergent by [Corollary 2.11](#) which is absurd.

If n is even, we have $P_Y x_n = e_1$ and so $x_n - P_Y x_n = (e_0 + e_1) - e_1 = e_0$; thus, $d_Y(x_n) = 1$. If n is odd, we have $x_n = e_1 + e_{n+1} \in Y$ and so $P_Y x_n = x_n = e_1 + e_{n+1} \rightharpoonup e_1$ and $d_Y(x_n) = 0$. Hence $P_Y x_n \rightharpoonup e_1$, which must be $A_Y(x_{\mathbb{N}})$ by [Theorem 3.12](#), and $(d_Y(x_n))_{n \in \mathbb{N}} = (1, 0, 1, 0, \dots)$. The fact that $(P_Y x_n)_{n \in \mathbb{N}}$ is not Opial with respect to Y follows from [Proposition 3.15\(i\)](#) below.

(ii): Because $C \subseteq Y$, it follows from (i) that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to C . By [5, Corollary 7.3], $P_C = P_B \circ P_Y$. Combing with (i) thus yields

$$P_C x_n = \begin{cases} e_1, & \text{if } n \text{ is even;} \\ \frac{1}{\sqrt{2}}(e_1 + e_{n+1}), & \text{if } n \text{ is odd.} \end{cases} \quad (20)$$

If n is even, then $x_n - P_C x_n = (e_0 + e_1) - e_1 = e_0$ and so $d_C(x_n) = 1$. If n is odd, then $x_n - P_C x_n = (1 - 1/\sqrt{2})(e_1 + e_{n+1})$ which yields $d_C(x_n) = \sqrt{2} - 1$. ■

Remark 3.14. Consider [Theorem 3.12](#). While $(P_C x_n)_{n \in \mathbb{N}}$ is weakly convergent, [Example 3.13\(i\)](#) illustrates that the sequence $(x_n - P_C x_n)_{n \in \mathbb{N}}$ may fail to converge weakly and the sequence $(d_C(x_n))_{n \in \mathbb{N}}$ may fail to converge as well. Furthermore, if we relax the assumption that C be affine to mere convexity, [Example 3.13\(ii\)](#) reveals that $(P_C x_n)_{n \in \mathbb{N}}$ may fail to be weakly convergent. Thus the conclusions in [Theorem 3.12](#) are sharp!

Proposition 3.15. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in X , and let Y be a closed linear subspace of X . Then the following hold:

- (i) Suppose that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to Y . Then: $(P_Y x_n)_{n \in \mathbb{N}}$ is Opial with respect to $Y \Leftrightarrow (d_Y(x_n))_{n \in \mathbb{N}}$ converges, in which case $(P_Y x_n)$ is weakly convergent.
- (ii) Suppose that $(x_n)_{n \in \mathbb{N}}$ is weakly convergent. Then: $(P_Y x_n)_{n \in \mathbb{N}}$ is Opial with respect to $Y \Leftrightarrow (\|P_Y x_n\|)_{n \in \mathbb{N}}$ is convergent.

Proof. (i): We have $(\forall y \in Y)(n \in \mathbb{N}) \|x_n - y\|^2 = \|P_Y x_n - y\|^2 + d_Y^2(x_n)$. Because $\lim_n \|x_n - y\|$ exists, the announced equivalences holds. In this case, [Corollary 2.11](#) gives the weak convergence of $(P_Y x_n)_{n \in \mathbb{N}}$ (in Y), which in turn gives the weak convergence of $(P_Y x_n)_{n \in \mathbb{N}}$ (in X).

(ii): Suppose that $x_n \rightharpoonup x \in X$, and write

$$(\forall y \in Y)(\forall n \in \mathbb{N}) \|P_Y x_n - y\|^2 = \|P_Y x_n\|^2 - \|x\|^2 + \|x - y\|^2 + 2 \langle x - P_Y x_n, y \rangle.$$

Note that $\langle x - P_Y x_n, y \rangle = \langle x - x_n, y \rangle \rightarrow 0$. Thus $(P_Y x_n)_{n \in \mathbb{N}}$ is Opial with respect to $Y \Leftrightarrow (\|P_Y x_n\|)_{n \in \mathbb{N}}$ is convergent. ■

Example 3.16. Suppose that $X = \mathbb{R}^2$ and set $Y := \mathbb{R} \times \{0\}$. Let $(\varepsilon_n)_{n \in \mathbb{N}}$ be a strictly decreasing sequence of positive real numbers with $\varepsilon_n \rightarrow 0$. Define the sequence $(x_n)_{n \in \mathbb{N}} := (\varepsilon_n, (-1)^n)_{n \in \mathbb{N}}$ in X . Then $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to Y because for every $y = (\eta, 0) \in Y$, we have $\|x_n - y\|^2 = (\varepsilon_n - \eta)^2 + 1 \rightarrow \eta^2 + 1$. There is no proper superset D of Y such that $(x_n)_{n \in \mathbb{N}}$ is Opial with respect to D . Moreover, $P_Y(x_n) = (\varepsilon_n, 0) \rightarrow (0, 0) \neq (\varepsilon_0, 0) = P_Y(x_0)$ and $(x_n - P_Y x_n)_{n \in \mathbb{N}} = (0, (-1)^n)_{n \in \mathbb{N}}$ does not converge.

Remark 3.17. If a sequence $(x_n)_{n \in \mathbb{N}}$ is Fejér monotone with respect to a closed linear subspace Y of X , then it is well known that $P_Y x_n \equiv P_Y x_0$ (see, e.g., [6, Proposition 5.9(i)]). [Example 3.16](#) illustrates that this projection property does not hold for Opial sequences.

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